# PROTOTYPE PROCESSOR FOR ARD OF COPERNICUS SAR MISSIONS
# ALGORITHM THEORETICAL BASELINE DOCUMENT (ATBD)


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|---|---|
| Contract Nº | 4000144699/24/I-NS |
| Reference | esa-rose-l-004-D4 |
| Issue/Revision | 2.2 |
| Date of Issue | 31/05/2026 |
| Status | Approved |

# APPROVAL

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|---|---|---|---|
| Title | Algorithm Theoretical Baseline Document (ATBD) | | |
| Issue Number | 2 | Revision Number | 2 |
| Author | F. Minati (B-Open), F. Trillo (B-Open), F. De Zan (Delta Phi), J.M. Lopez-Sanchez (Universidad de Alicante), P. Lopez Dekker (TU-Delft) | Date | 31/05/2026 |
| Approved by | | Date of Approval | |
| F. Minati | | 31/05/2026 | |

# CHANGE LOG

| Reason for change | Issue Nr. | Revision Number | Date |
|---|---|---|---|
| First issue | 1 | 0 | 15/05/2025 |
| Typos correction and answers to RIDs | 1 | 1 | 31/08/2025 |
| Answers to RIDs | 1 | 2 | 31/10/2025 |
| Document specific for CSLC/GSLC and CMLC/GMLC products | 2 | 0 | 31/10/2025 |
| Implement modifications requested at CDR meeting | 2 | 1 | 31/03/2026 |
| Implemented CRD-RID AV-XX | 2 | 2 | 31/05/2026 |

# CHANGE RECORD

| Issue Number 2 | Revision Number 2 | | |
|---|---|---|---|
| Reason for change | Date | Page(s) | Paragraph(s) |
| First version | 15/05/2025 | - | All |
| Minor typos correction | 31/08/2025 | - | All |
| Added jupyter notebooks for product generation and analysis (RIDs AV-33, AV-37) | 31/08/2025 | - | All |
| Added Detailed Variables Description tables for all section presenting algorithms (RID AV-33) | 31/10/2025 | - | All |
| ”Noise removal” section moved in the ”Flattening” chapter, together with ”Phase flattening” and ”Gamma flattening” sections (RID AV-38) | 31/10/2025 | 316 | 63.1 |
| Modified equation 65.12 in the ANF computation section (RID AV-13) | 31/10/2025 | 342 | 65.2.4 |
| Reduction of the chapters following the update to version 2.0 | 31/10/2025 | - | All |
| ”Noise removal” section moved in the ”Flattening” chapter, together with ”Phase flattening” and ”Gamma flattening” sections (RID AV-38) | 31/10/2025 | 150 | 21.1 |
| Modified equation 23.12 in the ANF computation section (RID AV-13) | 31/10/2025 | 178 | 23.2.4 |
| Modifications to the CMLP/GMLP SAOCOM demo | 31/03/2026 | 87 | 15 |
| Explanation of the reference image selection and update algorithm for CSLC product (RID AV-06) | 31/03/2026 | 189 | 24 |
| Clarification on the amplitude calibration operation for S-1 and ROSE-L missions (RID AV-39) | 31/03/2026 | 121 | 18.1.1 |
| Note on the resampling in the radar reference grid before data-driven co-registration (RID MP-02) | 31/03/2026 | 16 | 4 |
| Note on the gamma-flattening step operation (RID MP-03) | 31/03/2026 | 16 | 4 |
| Addition of the ”Interferometric phase and coherence generation, and phase unwrapping” Chapter | 31/03/2026 | 235 | 26 |
| Removal of the thermal noise removal operation in the amplitude calibration step | 31/03/2026 | 121 | 18 |
| Addition of some workflow diagrams | 31/03/2026 | - | All |
| Update of some of the existent workflow diagrams | 31/03/2026 | - | All |
| CMLC, GMLC and MLC terms substituted with CMLP, GMLP, MLP respectively (RID CA-01) | 31/03/2026 | - | All |
| Addition of the Split spectrum in the Data-driver Co-registration key steps | 31/03/2026 | 191 | 24.2 |
| Review of the Model-based and Data-driven Co-registration workflows | 31/03/2026 | 188 | 24 |
| Clarification onf the application of the phase geocoding step for CMLP products | 31/03/2026 | 119 | 17 |
| Review of the Reference geometry workflows | 31/03/2026 | 188 | 24 |
| Review of the Model-based and Data-driven Co-registration workflows | 31/03/2026 | 188 | 24 |
| Split spectrum section greatly enhanced, including code of core functions and additions in the Detailed Variables Description | 31/03/2026 | 206 | 24.5 |
| Clarifications on the ETAD corrections application in the image resampling step | 31/03/2026 | 142 | 20.1 |
| Modified structure of the document | 31/05/2026 | All | AI1 |